

**PaddyGuard AI: A Multimodal AI Based
System for Rice Leaf Disease & Pest
Detection, Treatment Recommendation
and Knowledge Assistance**

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Project Proposal Report
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Rice is one of the most important staple crops in many parts of the world, particularly in Asian countries where it plays a critical role in food security and rural livelihoods. However, rice production is frequently threatened by various diseases and pest infestations that significantly reduce crop yield and quality. Small-scale farmers often struggle to accurately identify these issues due to limited access to agricultural experts and diagnostic tools. As a result, pests and diseases are sometimes misidentified, leading to incorrect pesticide usage, economic loss, and environmental damage.

Although several artificial intelligence-based plant disease detection systems have been proposed in recent years, many existing solutions rely mainly on image-based classification and often lack the ability to handle uncertain or unknown inputs. Most systems assume that the input image always belongs to one of the trained categories, which can lead to unreliable predictions when unfamiliar insects or non-relevant images are provided. This limitation reduces the reliability of AI systems when deployed in real-world agricultural environments.

To address these challenges, this research proposes **PaddyGuard AI**, a multimodal intelligent decision support system designed for rice disease and pest management. The system integrates deep learning-based image analysis, voice-based symptom understanding, and a knowledge-driven advisory mechanism to support farmers in diagnosing crop health issues. In particular, the rice pest detection module utilizes convolutional neural network architectures such as ResNet, EfficientNet, and YOLO-based models to identify common rice pests from field images. To enhance reliability, Out-of-Distribution (OOD) detection techniques are incorporated to identify unknown insects that do not belong to the trained dataset.



Figure 1:pests imgs

The proposed system is expected to improve the accuracy and reliability of automated pest and disease detection while providing accessible and practical guidance for farmers. By supporting early detection and appropriate treatment recommendations, the system can contribute to improved crop productivity, reduced pesticide misuse, and more sustainable agricultural practices.

Keywords: Rice Pest Detection, Deep Learning, Out-of-Distribution Detection, Precision Agriculture, Computer Vision

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
CNN	Convolutional Neural Network
OOD	Out-of-Distribution
YOLO	You Only Look Once
NLP	Natural Language Processing
SDG	Sustainable Development Goals
GPU	Graphics Processing Unit
UI	User Interface
API	Application Programming Interface
ML	Machine Learning
WBS	Work Breakdown Structure
RAG	Retrieval Augmented Generation

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1. INTRODUCTION

1.1 Background and Context

Rice is one of the most widely consumed staple foods in the world and plays a vital role in ensuring food security, particularly in Asian countries such as Sri Lanka. Millions of smallholder farmers depend on rice cultivation as their primary source of income and livelihood. However, rice production is frequently threatened by various crop diseases and pest infestations, which can significantly reduce both yield and crop quality. In many agricultural regions, farmers often struggle to identify early symptoms of plant diseases or pest attacks due to limited access to expert agricultural advice and diagnostic tools.

Rice pests such as brown planthoppers, stem borers, and leaf folders are among the most common threats affecting paddy fields. These pests damage plant tissues, weaken crop growth, and can rapidly spread across large cultivation areas if not identified and managed early. According to global agricultural studies, pest infestations can cause up to 20–40% yield losses in rice production if proper management practices are not implemented in time. This problem particularly affects small-scale farmers who lack access to timely information and expert guidance. As a result, farmers may rely on trial-and-error approaches or excessive pesticide use, which can lead to increased production costs, environmental damage, and potential health risks.

In recent years, several digital agricultural advisory tools and mobile applications have been developed to assist farmers in identifying crop diseases using artificial intelligence and computer vision technologies. Systems such as Plantix and Agrio allow users to upload plant images and receive diagnostic suggestions. While these tools demonstrate the potential of AI in agriculture, many existing systems focus mainly on plant disease detection and often rely solely on image classification. A major limitation of these approaches is their inability to recognize unknown or unseen inputs. When an unfamiliar insect or irrelevant image is provided, the system may still produce a confident prediction even if the image does not belong to any known category. This limitation reduces the reliability and trustworthiness of automated diagnostic systems when applied in real-world farming environments.

To address these challenges, the proposed research introduces PaddyGuard AI, a multimodal intelligent decision support system designed to assist farmers in identifying rice diseases and pests and obtaining appropriate treatment guidance. The system integrates multiple artificial intelligence techniques, including deep learning-based image analysis, voice-based symptom understanding, and knowledge-based advisory support. Within this system, the rice pest detection module focuses on identifying common rice pests using deep learning models such as ResNet, EfficientNet, and YOLO-based detection architectures. To improve reliability, the system incorporates Out-of-Distribution (OOD) detection techniques that can identify unfamiliar insect images that do not belong to the trained dataset. This capability helps prevent incorrect predictions and enhances the robustness of the overall decision support system.

The proposed research aligns with the global Sustainable Development Goals, particularly SDG 2 (Zero Hunger), which focuses on improving food security and sustainable agriculture. By enabling early detection of crop threats and supporting informed decision-making for pest management, the system has the potential to improve agricultural productivity and reduce crop losses. In addition, the research contributes to the broader field of artificial intelligence applications in agriculture, combining techniques from computer vision, natural language processing, and intelligent decision support systems. Within the academic context, the study aligns with research clusters related to Artificial Intelligence, Machine Learning, and Agricultural Informatics, demonstrating the interdisciplinary nature of modern agricultural technology solutions.

1.2 Literature Review

Recent advancements in artificial intelligence and computer vision have significantly improved automated crop monitoring and pest detection systems. Deep learning techniques, particularly convolutional neural networks (CNNs), have demonstrated strong performance in image-based agricultural diagnostics. Several studies have explored the use of deep learning architectures for detecting rice diseases and insect pests with the aim of supporting farmers in early crop protection and yield improvement. This section reviews key scholarly contributions related to deep learning-based pest detection, mobile agricultural diagnostic systems, and reliability improvements through uncertainty detection techniques.

1.2.1 Deep Learning for Rice Pest Detection

Recent studies have demonstrated that deep learning models can effectively identify rice pests from agricultural images. Hu et al. (2023) investigated the application of convolutional neural networks for rice pest recognition using large-scale annotated image datasets. Their findings showed that CNN-based models can automatically extract complex visual features of insect pests and achieve significantly higher classification accuracy compared to traditional machine learning techniques. However, the authors highlighted that visual similarity between pest species and complex field backgrounds can introduce classification challenges in real-world environments.

Similarly, Zheng et al. (2024) proposed a lightweight rice pest detection framework based on an improved YOLOv8 architecture. Their model demonstrated strong performance in detecting small insect objects within field images while maintaining computational efficiency suitable for real-time applications. Nevertheless, the study noted that pest detection accuracy may be affected by factors such as varying lighting conditions, occlusions, and small object sizes that are common in agricultural environments.

1.2.2 Deep Learning for Crop Disease Diagnosis

In addition to pest detection, several studies have explored the use of deep learning models for automated crop disease diagnosis. Pai et al. (2025) developed a rice leaf disease detection system using convolutional neural networks trained on large-scale annotated datasets. The study demonstrated that CNN-based classifiers can successfully identify multiple rice disease categories with high accuracy under controlled experimental conditions. However, the authors observed that model performance may decrease when images are captured in uncontrolled field environments where lighting conditions, background noise, and plant variations are present.

Similarly, Deng et al. (2023) investigated mobile-based deep learning solutions for detecting rice diseases and pests using smartphone applications. Their work showed that lightweight deep learning models can be deployed on mobile devices to assist farmers in real-time crop monitoring. Despite these advantages, the authors pointed out that many existing systems rely on closed-set classification assumptions, where the model is forced to classify every input image into one of the predefined categories.

1.2.3 Machine Learning in Agricultural Pest and Disease Detection

More broadly, several review studies have examined the growing application of machine learning and deep learning techniques in agricultural pest and disease detection. Wang et al. (2025) conducted a comprehensive review of deep learning approaches used in agricultural diagnostics and concluded that these models significantly outperform traditional image processing techniques in terms of accuracy and scalability.

However, the review also emphasized that many AI-based agricultural systems struggle to generalize across diverse environmental conditions. Variations in lighting, crop growth stages, and background complexity often reduce the reliability of models trained on controlled datasets.

1.2.4 Out-of-Distribution Detection for Reliable AI Systems

Another important research direction focuses on improving model reliability through uncertainty detection techniques. Cui et al. (2022) explored several methods for Out-of-Distribution (OOD) detection in deep neural networks. OOD detection techniques enable AI models to recognize inputs that fall outside the training data distribution, preventing models from generating highly confident but incorrect predictions.

Similarly, Hsu et al. (2020) introduced the Generalized ODIN method for detecting out-of-distribution samples without requiring explicit training on unknown classes. Their approach demonstrated improved detection performance across multiple image classification tasks. Integrating such techniques into agricultural diagnostic systems can significantly enhance reliability when the system encounters unfamiliar insect species or irrelevant images.

1.2.5 Limitations of Existing Systems

Although existing research demonstrates the effectiveness of deep learning for agricultural pest and disease detection, several limitations remain. Many current systems rely heavily on supervised learning models trained on predefined datasets and assume that every input image belongs to one of the known classes. In practical agricultural environments, however, farmers may capture images containing unknown insect species, multiple pests within a single image, or irrelevant objects.

Furthermore, models trained in controlled environments often struggle when applied to real-world field conditions where lighting, background complexity, and environmental variability differ significantly from the training dataset. These limitations highlight the need for more robust pest detection systems that can both identify known pest species and recognize unfamiliar inputs.

1.3 Research Gap

A critical analysis of existing research in deep learning-based agricultural pest detection reveals several important limitations that remain insufficiently addressed in current systems. The following research gaps form the basis for the proposed study:

- **Closed-set classification limitation:**

Most existing pest detection systems are trained using supervised learning models that assume every input image belongs to one of the predefined pest categories. In real agricultural environments, however, farmers may capture images containing unfamiliar insect species or irrelevant objects. Current systems lack mechanisms to identify such unknown inputs and therefore may produce highly confident but incorrect predictions.

- **Limited capability for detecting unknown pest species:**

Existing AI-based agricultural diagnostic systems are typically designed to recognize only the pest classes included in their training datasets. As a result, these systems cannot effectively handle images of new or unseen insect species that may appear in real farming environments. This limitation reduces the reliability and practical usability of automated pest detection systems.

- **Performance degradation under real-world field conditions:**

Many deep learning models demonstrate high accuracy when evaluated using controlled datasets. However, their performance often decreases when applied to real-world agricultural environments where images may contain complex backgrounds, varying lighting conditions, occlusions, and multiple insects within a single image.

- **Lack of uncertainty-aware prediction mechanisms:**

Most current pest detection models do not incorporate techniques to handle prediction

uncertainty. Without such mechanisms, models may incorrectly classify unfamiliar inputs rather than recognizing that the input does not belong to the known training distribution.

• **Limited integration of Out-of-Distribution detection in agricultural AI systems:** Although Out-of-Distribution (OOD) detection has been widely studied in general machine learning research, its application in agricultural pest detection systems remains limited. Integrating OOD detection into pest recognition models can significantly improve system reliability by enabling the detection of unknown insect samples that fall outside the training data distribution.

This research addresses these limitations by developing a deep learning–based rice pest detection module that integrates Out-of-Distribution detection techniques within the proposed PaddyGuard AI system. By enabling the model to both classify known pests and identify unfamiliar insect samples, the proposed approach aims to improve the robustness and reliability of AI-based pest detection systems in real-world agricultural environments.

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design and develop a deep learning–based rice pest detection module capable of accurately identifying common rice pests from field images while improving prediction reliability through the integration of Out-of-Distribution (OOD) detection techniques within the PaddyGuard AI decision support system.

2.2 Specific Objectives

- To collect and prepare a representative dataset of rice pest images from publicly available agricultural datasets and other reliable sources for training and evaluation.
- To develop and train deep learning models, such as Convolutional Neural Network (CNN) architectures and YOLO-based detection models, for accurate identification of common rice pests from image data.
- To apply image preprocessing and data augmentation techniques in order to improve model generalization under varying environmental conditions such as lighting variations, background complexity, and different pest appearances.
- To implement Out-of-Distribution (OOD) detection mechanisms that enable the system to identify insect images that do not belong to the trained pest categories, thereby reducing incorrect predictions.
- To evaluate the performance and reliability of the developed pest detection model using standard evaluation metrics such as accuracy, precision, recall, and F1-score under both controlled and realistic image conditions.

- To integrate the developed rice pest detection module into the PaddyGuard AI system to support intelligent crop monitoring and assist farmers in identifying potential pest infestations at an early stage.

3. METHODOLOGY

3.1 Research Approach

This research adopts a **Design Science and Experimental research approach** to develop and evaluate a reliable rice pest detection module within the PaddyGuard AI system. Design Science Research is appropriate for this study because the objective is to design and implement a technological artifact that addresses a real-world agricultural problem. The proposed artifact in this research is a deep learning–based pest detection module capable of identifying rice pests and detecting unknown insect samples.

The experimental aspect of the research focuses on training and evaluating deep learning models using image datasets containing rice pest images. Different model architectures and detection strategies will be evaluated to determine the most reliable approach for identifying pests under varying environmental conditions. The overall research process includes dataset preparation, model development, performance evaluation, and integration of the detection module into the PaddyGuard AI decision support system.

3.2 Technical Methods and Algorithms

The technical implementation of the proposed system is based on **deep learning and computer vision techniques**. Convolutional Neural Networks (CNNs) will be used as the primary method for feature extraction and image classification. CNN architectures such as **ResNet50** and **EfficientNet** will be explored due to their strong performance in image recognition tasks and their ability to learn complex visual patterns from image data.

In addition to image classification, **object detection techniques** will also be considered for identifying pest instances within images. Models such as **YOLO-based architectures** can detect insects even when they appear in cluttered backgrounds or multiple insects are present in a single image. These detection models are well suited for agricultural image analysis where pests may occupy small regions within a large image.

To improve model robustness, **image preprocessing and data augmentation techniques** will be applied before training the deep learning models. Image preprocessing includes resizing, normalization, and noise reduction to standardize the input images. Data augmentation techniques such as image rotation, horizontal flipping, scaling, and brightness adjustments will be applied to artificially expand the training dataset and improve the model's ability to generalize across different field conditions.

A key component of this research is the integration of **Out-of-Distribution (OOD) detection techniques**. Traditional classification models assume that every input belongs to one of the

trained classes. However, in real-world scenarios farmers may capture images containing unknown insects or irrelevant objects. To address this challenge, OOD detection techniques such as **ODIN or Mahalanobis distance–based detection** will be used to identify samples that do not belong to the training distribution. This mechanism improves the reliability of the pest detection system by preventing incorrect predictions when unfamiliar insect species appear.

3.3 Tools and Platforms

The proposed system will be implemented using widely adopted tools and platforms for artificial intelligence and machine learning development. **Python** will be used as the primary programming language due to its extensive ecosystem of libraries for data science, machine learning, and image processing.

Deep learning models will be implemented using frameworks such as **PyTorch or TensorFlow**, which provide efficient tools for developing convolutional neural networks and training large-scale machine learning models. These frameworks also support GPU acceleration, which significantly improves training performance for deep learning tasks.

Image preprocessing and data augmentation operations will be performed using libraries such as **OpenCV, NumPy, and Matplotlib**. These libraries provide powerful functionalities for image manipulation, visualization, and numerical computation required for computer vision research.

For model development and experimentation, environments such as **Jupyter Notebook or Google Colab** may be used, as they provide convenient platforms for conducting experiments, visualizing results, and managing datasets during the research process.

3.4 Data Collection and Ethical Considerations

The dataset used for this research will consist of rice pest images collected from publicly available agricultural datasets and research repositories. These datasets typically include images of common rice pests such as **brown planthoppers, stem borers, and leaf folders**, which are known to cause significant damage to rice crops.

The collected dataset will be organized into three subsets: **training, validation, and testing datasets**, following standard machine learning practices. The training dataset will be used to train the deep learning models, while the validation dataset will assist in tuning model parameters. The testing dataset will be used to evaluate the final performance of the trained model.

Ethical considerations will also be addressed during the research process. Only publicly available datasets or properly attributed image sources will be used to ensure compliance with data usage policies. In addition, the research will focus solely on agricultural image data and will not involve any personal or sensitive information.

3.5 Validation Strategy and Performance Metrics

To evaluate the effectiveness of the proposed rice pest detection model, a validation strategy based on standard machine learning evaluation techniques will be used. The dataset will be divided into training, validation, and testing sets to ensure fair and reliable model evaluation.

The performance of the deep learning models will be assessed using commonly used classification metrics including **accuracy, precision, recall, and F1-score**. These metrics provide a comprehensive evaluation of model performance by measuring both prediction correctness and the ability of the model to correctly identify pest categories.

In addition, **confusion matrices** will be used to analyze classification performance across different pest categories and identify potential misclassification patterns. The effectiveness of the OOD detection mechanism will also be evaluated by testing the model with images that do not belong to the trained pest classes. This evaluation will ensure that the system can correctly identify unknown inputs and avoid generating unreliable predictions.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed System

The proposed PaddyGuard AI system is designed as a multimodal intelligent decision support platform that assists farmers in identifying rice crop diseases and pest infestations. The system integrates several artificial intelligence components including image-based disease detection, pest detection, voice-based symptom analysis, and a knowledge-driven advisory module. These components work together to provide reliable crop health diagnostics and practical recommendations for farmers.

The overall architecture of the system follows a modular design where each component performs a specific task within the crop monitoring pipeline. Farmers can interact with the system through a mobile or web interface by providing input data such as images of rice leaves or pests, or voice descriptions of observed crop symptoms. The system processes this information using AI models and provides diagnostic results along with treatment guidance.

Within this architecture, the rice pest detection module plays an important role in identifying insect pests that may affect rice crops. The module analyzes uploaded pest images using deep learning models and determines the likely pest category. In addition, an Out-of-Distribution (OOD) detection mechanism is integrated into this component to identify unfamiliar insect images that are not part of the trained dataset.

4.2 System Components

The high-level architecture of PaddyGuard AI consists of several interconnected components that collectively support crop health monitoring and decision making.

The **input layer** collects user data including rice leaf images, pest images, and voice-based symptom descriptions provided by farmers. These inputs are then processed by the **data preprocessing layer**, where image normalization, resizing, and augmentation operations are applied to prepare the data for analysis.

The **AI inference layer** contains the main machine learning models used for diagnosis. This layer includes the rice disease detection module, the rice pest detection module, and the voice-based symptom classification module. Each module analyzes its respective input and produces predictions related to crop health conditions.

The **decision support layer** integrates the outputs from the AI models and provides meaningful feedback to the user. This layer may include rule-based validation mechanisms, treatment recommendations, and knowledge retrieval systems that assist farmers in selecting appropriate crop management strategies.

Finally, the **output layer** communicates the diagnostic results to the user in a clear and accessible format, which may include text-based explanations, confidence scores, and voice responses.

4.3 Rice Pest Detection Component

The rice pest detection module is one of the key components of the PaddyGuard AI system. This component focuses on identifying common rice pests from images captured in agricultural fields. Deep learning models such as Convolutional Neural Networks (CNNs) are used to extract visual features from pest images and classify them into predefined pest categories.

In addition to standard classification capabilities, the module integrates Out-of-Distribution (OOD) detection mechanisms. These techniques enable the system to identify input images that do not belong to the trained pest classes. When an unknown insect image is detected, the system can notify the user that the input does not match known pest categories instead of generating an incorrect prediction. This improves the reliability and trustworthiness of the pest detection system in real-world farming environments.

4.4 Component Interaction and Data Flow

The interaction between system components follows a sequential data processing pipeline. When a farmer uploads an image of a suspected pest, the image first passes through the preprocessing stage where it is resized and normalized. The processed image is then forwarded to the pest detection model within the AI inference layer.

The trained deep learning model analyzes the image and produces a prediction indicating the most likely pest category. Simultaneously, the OOD detection mechanism evaluates whether the image belongs to the known training distribution. If the image is identified as an unknown sample, the system generates a warning instead of returning an incorrect classification.

The prediction results are then forwarded to the decision support module where additional contextual information and advisory recommendations can be generated. Finally, the processed information is delivered to the user through the system interface.

4.5 Scalability and Security Considerations

The proposed system architecture is designed to support scalability and secure handling of agricultural data. Cloud-based infrastructure can be used to host the AI models and manage computational resources required for deep learning inference. This enables the system to support multiple users simultaneously while maintaining efficient response times.

From a security perspective, the system will ensure that user data such as uploaded images or voice inputs are handled responsibly. Only necessary data will be processed for crop diagnosis purposes, and no personal or sensitive information will be collected during system usage. Additionally, secure communication protocols can be implemented to protect data transmission between the user interface and the AI processing components.

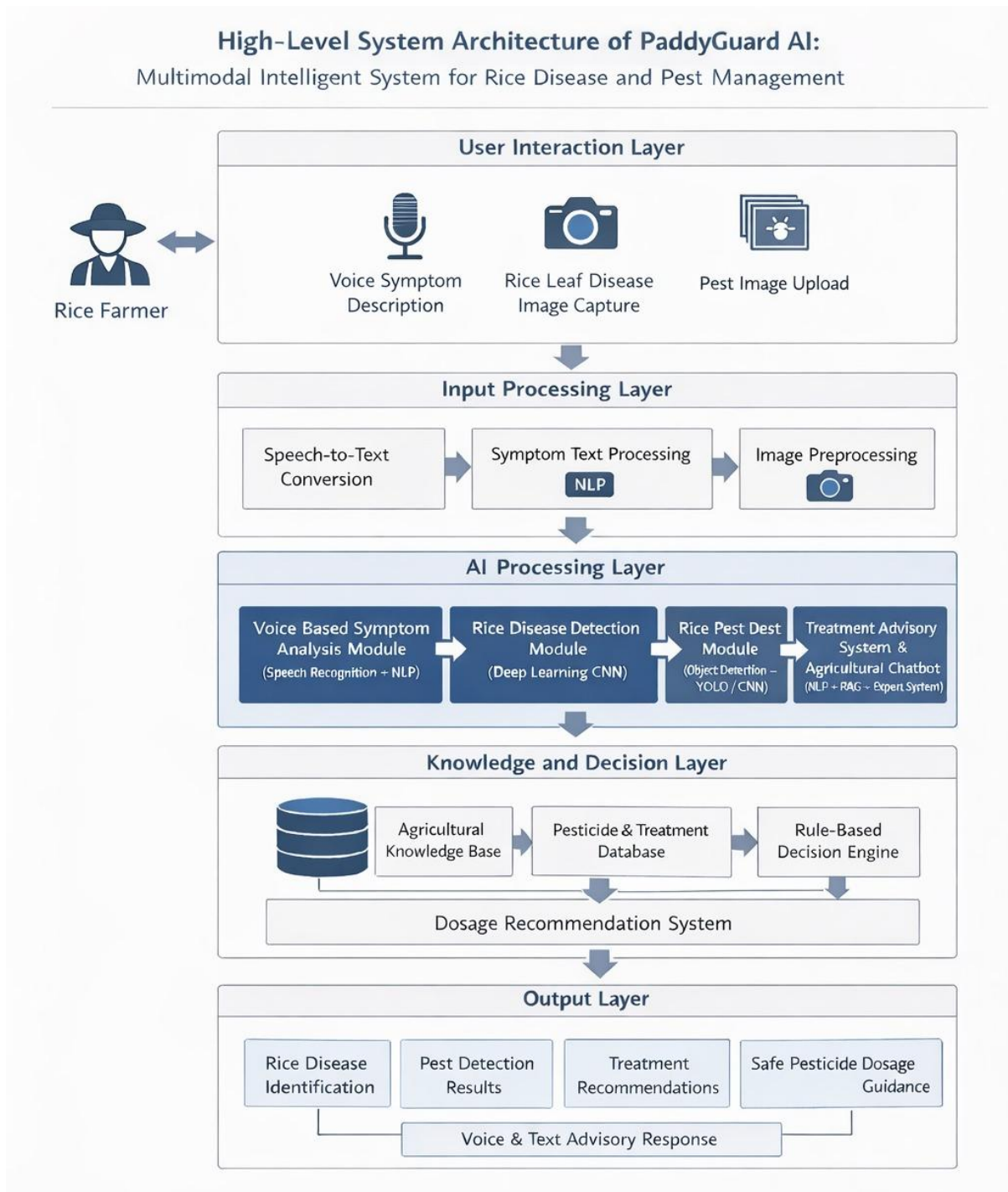


Figure 2: High Level System Architecture

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The PaddyGuard AI system is designed to support multiple stakeholders involved in rice cultivation, agricultural advisory services, and crop monitoring activities. These stakeholders can be categorized into three main groups based on their level of interaction with the system.

- **Primary Stakeholders:**

The primary stakeholders are rice farmers, particularly smallholder farmers who rely on paddy cultivation as their primary source of livelihood. Many farmers face difficulties in accurately identifying pest infestations due to limited access to agricultural experts and diagnostic tools. The proposed system provides these farmers with a simple and accessible tool to upload pest images and receive diagnostic feedback, enabling them to take timely actions to protect their crops.

- **Secondary Stakeholders:**

Agricultural extension officers, crop advisors, and agronomists represent the secondary stakeholder group. These professionals can use the system as a supporting decision-making tool when assisting farmers in identifying pest infestations and recommending appropriate crop management strategies. The system can also help extension officers provide faster and more consistent advisory services during field visits.

- **Tertiary Stakeholders:**

Government agricultural institutions such as the Department of Agriculture, agricultural research organizations, and agricultural technology developers form the tertiary stakeholder group. These stakeholders may benefit from aggregated system data to analyze pest occurrence patterns, monitor crop health trends, and improve future agricultural technologies and policy decisions related to crop protection and food security.

5.2 Functional Requirements

FR1: The system shall allow farmers to upload images of suspected rice pests captured in their fields.

FR2: The system shall preprocess uploaded images using resizing, normalization, and other image preparation techniques before analysis.

FR3: The system shall analyze pest images using trained deep learning models to identify the most likely pest category.

FR4: The system shall implement Out-of-Distribution (OOD) detection to recognize insect images that do not belong to the trained pest classes.

FR5: The system shall display the predicted pest category together with a confidence score.

FR6: The system shall alert the user when an uploaded image is detected as an unknown insect instead of providing an incorrect classification.

FR7: The system shall provide diagnostic results that can assist farmers in understanding pest infestations affecting rice crops.

5.3 Non-Functional Requirements

Performance:

The system should provide pest detection results within a short response time so that farmers can receive timely guidance.

Accuracy:

The detection model should achieve high accuracy when identifying common rice pests under different field conditions.

Usability:

The interface should be simple and easy to use, allowing farmers with limited technical knowledge to interact with the system.

Scalability:

The system architecture should support increasing numbers of users and image inputs without performance degradation.

Security:

Uploaded images and system data should be handled securely to prevent unauthorized access.

Reliability:

The system should provide consistent and dependable results during repeated usage in different environmental conditions.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the PaddyGuard AI system includes rice farmers who require practical and affordable technological support to identify crop diseases and pest infestations. In many agricultural regions, particularly in developing countries, farmers often lack immediate access to agricultural experts or diagnostic tools. As a result, early detection of crop threats becomes difficult, which can lead to reduced crop yields and financial losses.

The typical customer persona for the system is a small- to medium-scale rice farmer who uses a smartphone to capture images of crops and seeks quick diagnostic guidance. Agricultural extension officers and crop advisors also represent an important secondary user group, as they can use the system to assist farmers in diagnosing crop health issues more efficiently.

6.2 Value Proposition

PaddyGuard AI offers a practical solution for early detection of rice diseases and pest infestations using artificial intelligence. By combining image-based pest detection, disease identification, and intelligent advisory support, the system helps farmers make informed decisions about crop management.

The key value of the proposed system lies in its ability to provide fast and reliable crop diagnostics directly from field images. In addition, the integration of Out-of-Distribution detection techniques improves the reliability of predictions by identifying unknown insect samples instead of generating incorrect classifications. This feature enhances user trust and increases the practical usability of the system.

6.3 Revenue Model

The system can adopt multiple revenue models depending on the deployment environment. One potential approach is a **freemium model**, where basic crop diagnostic services are offered free of charge to farmers while advanced advisory features are provided through a subscription plan.

Another possible revenue model involves partnerships with agricultural organizations, government institutions, or agricultural technology companies. These organizations may adopt the system as part of larger agricultural monitoring programs, enabling wider distribution of the technology.

6.4 Cost Estimation

The main cost components involved in developing and deploying the PaddyGuard AI system include software development, dataset preparation, cloud infrastructure, and system maintenance. Additional costs may arise from training deep learning models and managing computational resources required for model inference.

Cloud-based infrastructure may be used to host the trained models and manage large-scale image processing tasks. While cloud services introduce operational costs, they also provide scalability and reliability for supporting multiple users simultaneously.

6.5 Pricing Strategy

A flexible pricing strategy can be adopted to make the system accessible to farmers with different financial capabilities. Basic diagnostic features can remain freely accessible to encourage adoption among farmers, while premium features such as advanced analytics or large-scale monitoring tools may be offered through subscription plans.

Government-supported agricultural programs could also subsidize the system to ensure that smallholder farmers can benefit from the technology without significant financial burden.

6.6 Competitive Advantage

The PaddyGuard AI system offers several advantages compared to existing crop diagnostic applications. Many existing systems rely solely on image classification for plant disease detection and may generate incorrect predictions when unfamiliar inputs are provided.

In contrast, the proposed system integrates **Out-of-Distribution detection**, which improves prediction reliability by identifying unknown insect samples. Additionally, the multimodal design of the system allows it to combine image analysis with voice-based symptom descriptions and knowledge-driven advisory support. These features create a more comprehensive agricultural decision support platform.

6.7 Intellectual Property Considerations

The technological components of the PaddyGuard AI system, including the trained machine learning models, system architecture, and advisory mechanisms, may represent valuable intellectual property. Appropriate measures can be taken to protect these innovations through software licensing, copyrights, or patents where applicable.

In addition, the system will respect the intellectual property rights associated with publicly available datasets and research resources used during development. Proper attribution and compliance with dataset usage policies will be maintained throughout the research and deployment process.

7. BUDGET AND JUSTIFICATION

7.1 Estimated Project Budget

This section presents the estimated budget required for the development and evaluation of the proposed **PaddyGuard AI system**. Since this project is conducted as an academic research project, the overall cost is minimized by utilizing open-source software tools, existing personal hardware resources, and free or low-cost cloud services where possible. Additional costs are estimated only for essential resources required for machine learning experimentation, dataset storage, and system development.

7.2 Hardware Costs

The research will primarily be conducted using an existing personal laptop owned by the researcher. Therefore, no additional hardware purchases are required. However, a small allocation is included for optional external storage to securely store pest image datasets and trained model outputs.

Table 1: Hardware Costs

Hardware	Purpose	Estimated Cost (LKR)
External Storage (Optional)	Backup of datasets and model outputs	Rs. 5,000
Total Cost		Rs. 5,000

The use of an existing personal laptop significantly reduces hardware costs while still providing the necessary computational resources for experimentation and development.

7.3 Software Licenses

The development of the PaddyGuard AI system relies mainly on open-source software tools widely used in artificial intelligence research and machine learning development. These tools provide powerful development capabilities without requiring expensive licensing fees.

Table 2: Software Licenses

Software	Purpose	License Type	Cost (LKR)
Python 3.x	Programming language for AI model development	Open Source	0
PyTorch / TensorFlow	Deep learning framework for model training	Open Source	0
OpenCV	Image processing and computer vision tasks	Open Source	0
NumPy / Pandas	Data processing and numerical computations	Open Source	0
VS Code	Integrated development environment	Free	0
TOTAL			0

The use of open-source technologies significantly reduces software licensing expenses while still providing access to advanced machine learning frameworks required for the project.

7.4 Cloud Services

Cloud computing services may be used to support large-scale model training, dataset storage, and system experimentation. Cloud platforms provide GPU resources that accelerate deep learning model training and allow efficient management of large image datasets.

Table 3: Cloud Services

Service	Usage	Plan	Cost (LKR)
Google Colab Pro	GPU-accelerated model training	Monthly subscription	15,000
Cloud Storage (AWS S3 / Google Cloud Storage)	Dataset storage, model checkpoints, and API hosting during prototype phase	Pay-as-you-go	6,000
Domain & SSL Certificate	Secure HTTPS API endpoint for prototype deployment	Annual	3,500
TOTAL			24,500

Cloud-based GPU resources significantly reduce the need for expensive hardware upgrades and allow efficient training of deep learning models used for pest detection.

7.5 Human Effort Cost

Since this research project is conducted as part of an undergraduate academic study, no direct labor costs are included. However, a small allocation is included for report preparation and submission requirements.

Table 4: Human Effort Cost

Role	Task	Cost (LKR)
Printing & Binding	Final research report and documentation submission	3,000
TOTAL		3,000

7.6 Justification of Expenses

Table 5: Justification of Expenses

Category	Total Cost (LKR)
Hardware Costs	5,000
Software Licenses	0
Cloud Services	24,500
Human Effort Cost	3,000
GRAND TOTAL	32,500 LKR

The estimated budget reflects the essential resources required to successfully develop and evaluate the **PaddyGuard AI system**. Cloud-based GPU services enable efficient training of deep learning models such as ResNet50, EfficientNet, and YOLOv8 without requiring expensive hardware upgrades. Dataset storage and secure API deployment are supported through cloud storage services and domain hosting.

The use of open-source software technologies significantly reduces licensing costs while still providing access to powerful machine learning frameworks. Overall, the proposed budget ensures that the project remains **cost-effective, technically feasible, and suitable for an academic research environment** while supporting the development of the AI-based pest detection system.

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) presents the major tasks involved in the development of the PaddyGuard AI system and allocates responsibilities among the project team members. The tasks are organized to ensure balanced workload distribution while aligning with the overall twelve-month research timeline. Each member is responsible for developing a specific module of the system while also collaborating on common activities such as dataset preparation, system integration, evaluation, and documentation.

Table 6: WBS

	Task Description	Responsible Member	Timeline
T1	Literature review and research problem analysis	All members	Month 1–2
T2	Dataset collection and preprocessing	All members	Month 2–3
T3	Rice disease detection model development	Member 1	Month 3–7
T4	Rice pest detection model development	Member 2	Month 3–7

	Task Description	Responsible Member	Timeline
T5	Voice-based symptom analysis module develop	Member 3	Month 4–8
T6	Treatment advisory and knowledge system development	Member 4	Month 5–8
T7	Integration of system modules	All members	Month 8–9
T8	System testing and performance evaluation	All members	Month 9–10
T9	System refinement and optimization	All members	Month 10–11
T10	Final documentation and research report preparation	All members	Month 11–12

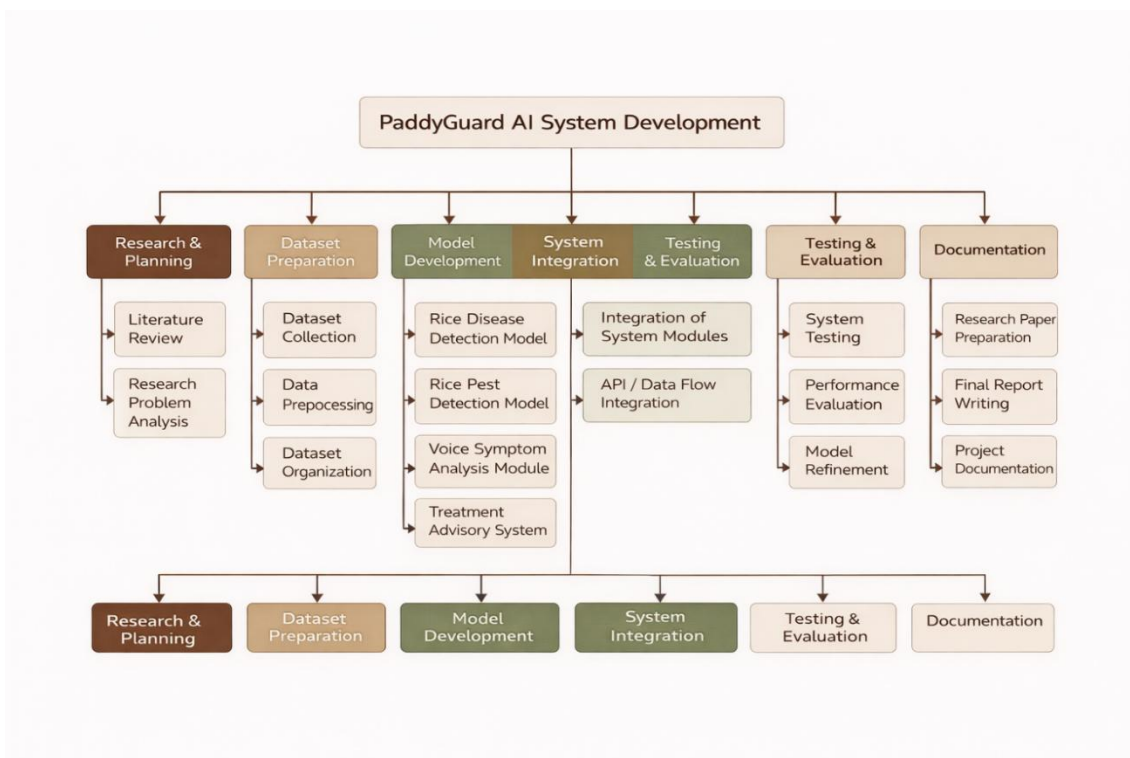


Figure 3 : WBS

The proposed task allocation ensures that each team member focuses on a specific technical component while maintaining collaborative involvement during the integration, testing, and documentation stages of the project. This structured approach supports efficient project management and helps ensure that all research objectives are achieved within the planned twelve-month timeline.

9. GANTT CHART

The Gantt chart presents the overall timeline for the development of the PaddyGuard AI system over a twelve-month research period. The project is divided into several phases including literature review, dataset preparation, model development, system integration, testing, and final documentation. Each phase is scheduled to ensure that the research activities progress in a logical sequence while allowing certain components to be developed in parallel.

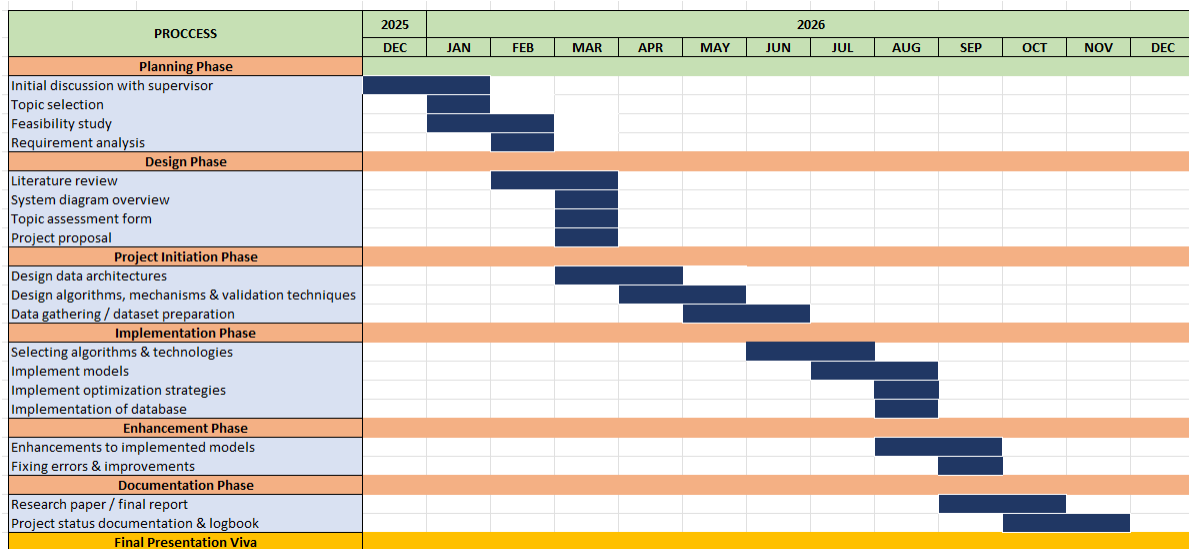


Figure 4: Gantt Chart

The timeline allows multiple modules of the PaddyGuard AI system to be developed simultaneously while maintaining coordination during integration and evaluation stages. This schedule ensures that sufficient time is allocated for model development, testing, system refinement, and preparation of the final research report.

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11. APPENDICES

The appendices provide additional supporting materials related to the development of the PaddyGuard AI system. These materials include supplementary diagrams, system interface concepts, risk analysis, and other documentation that supports the research process and system design.

Appendix A – System Architecture Diagram

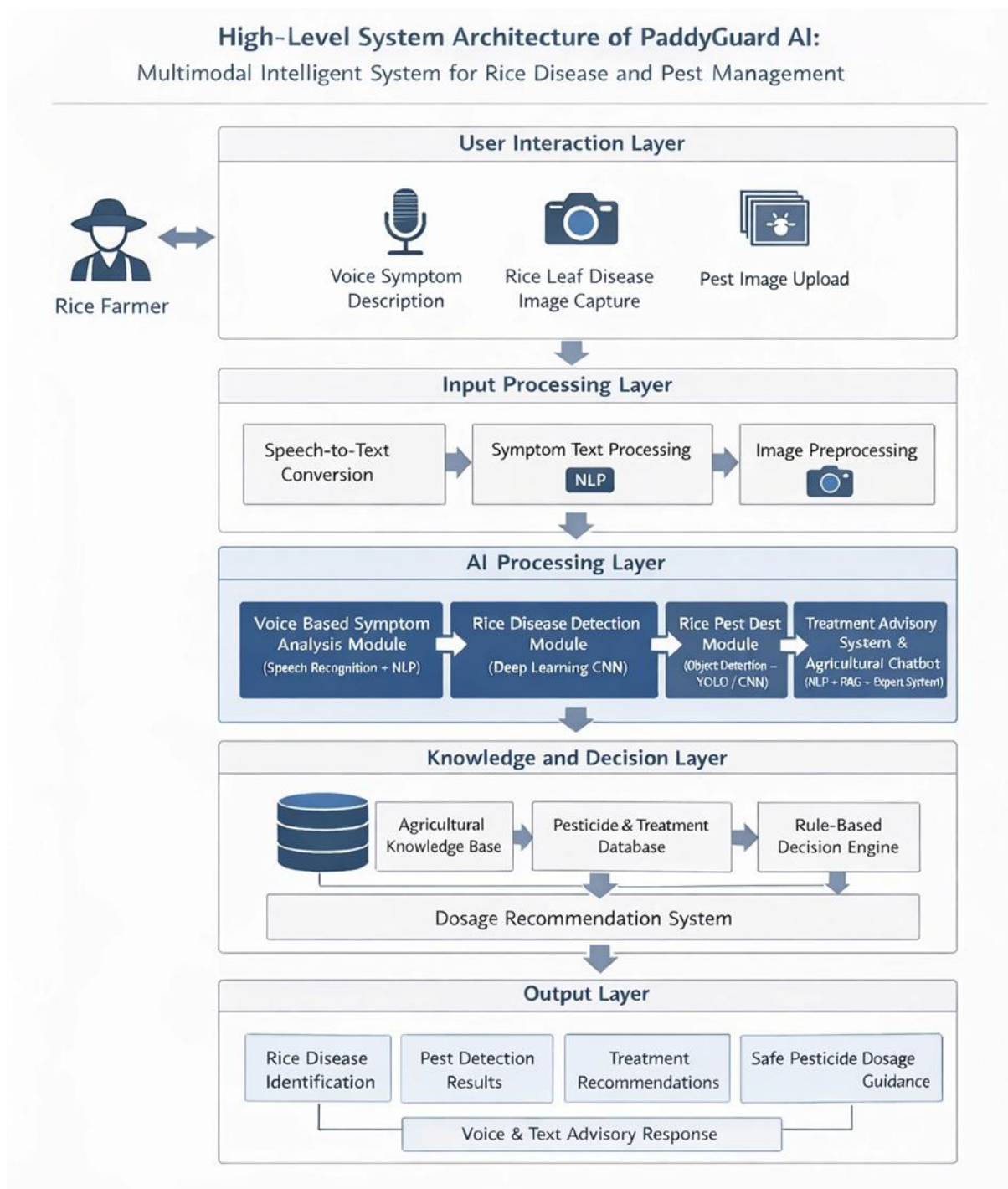


Figure 5: System Architecture Diagram

Appendix B – Sample Rice Pest Image Dataset

IP102 – Large Scale Insect Pest Dataset

<https://github.com/xpwu95/IP102>

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Appendix C – Preliminary User Interface Mockups

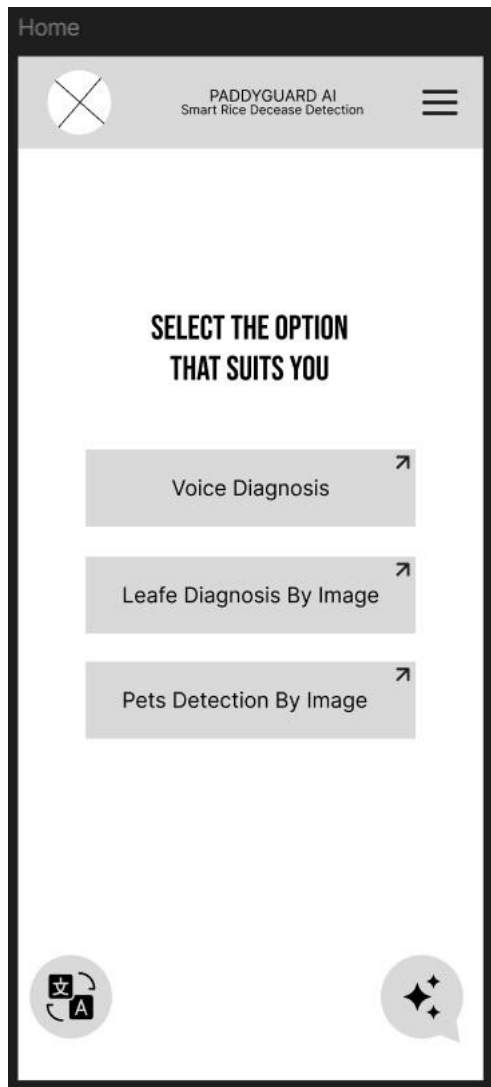


Figure 6: Home Page

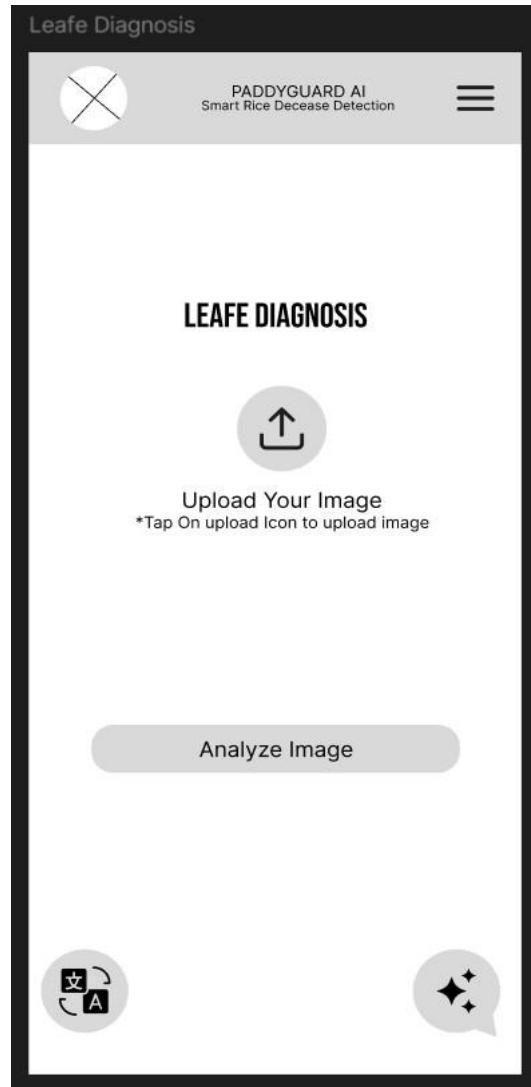


Figure 7: Leafe Diagnosis



Figure 8: Pets Detection

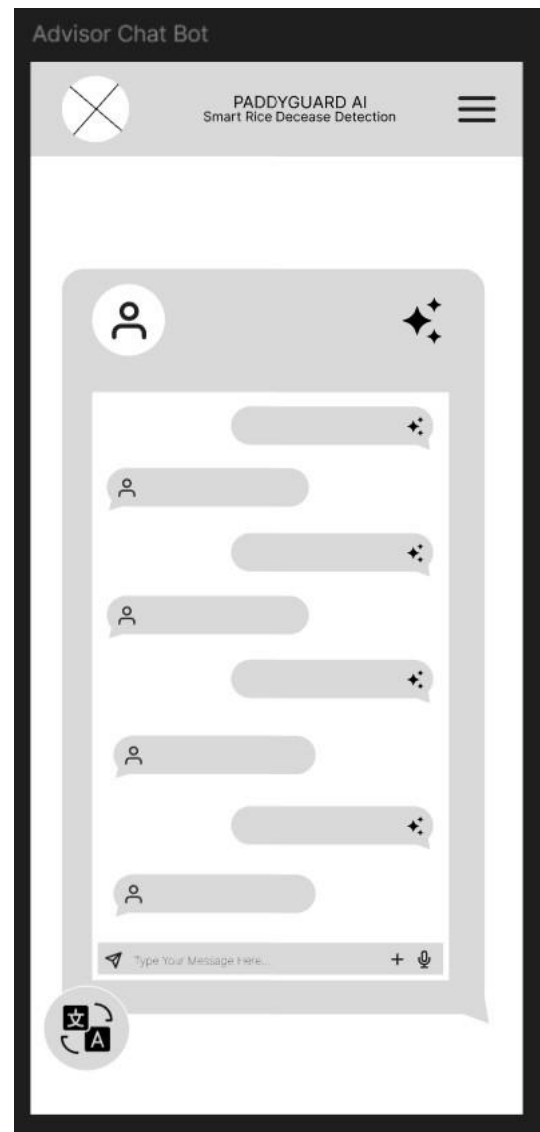


Figure 9: Chat Bot

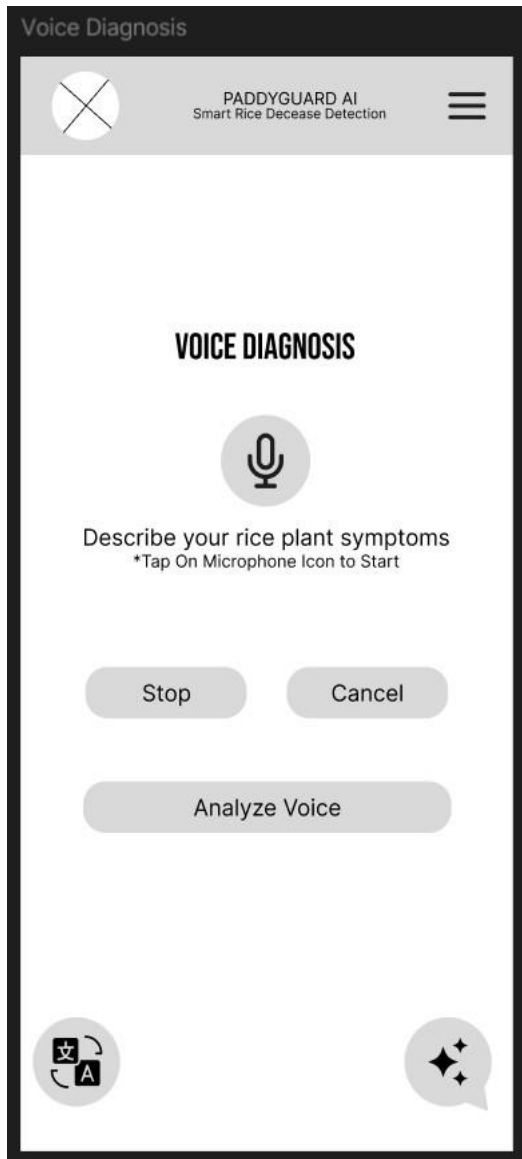


Figure 10: Voice Diagnosis

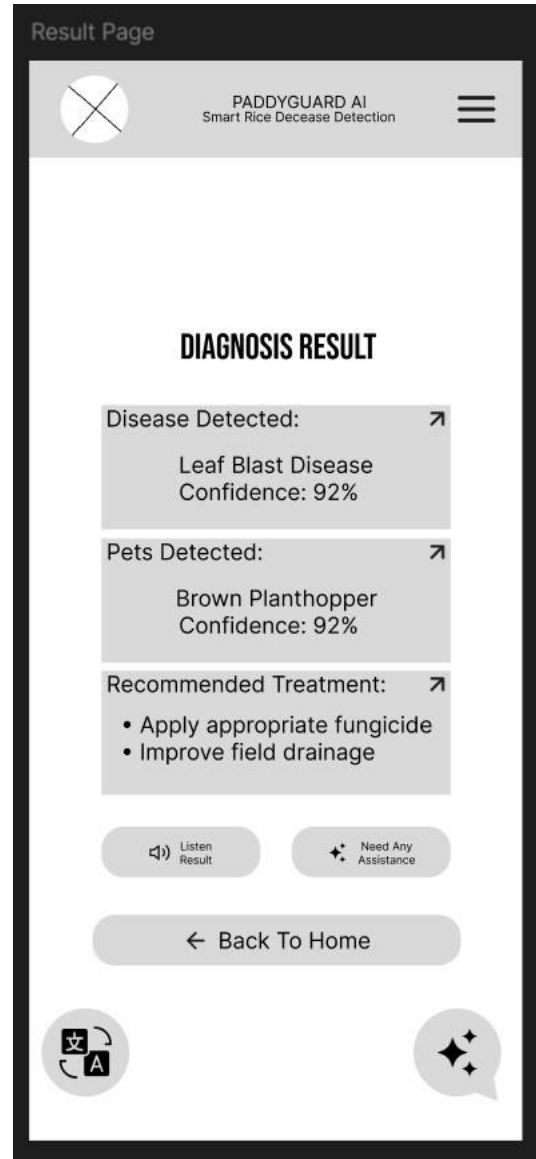


Figure 11: Result Page

Appendix D – Project Risk Analysis

This appendix summarizes potential risks that may affect the development of the system and outlines mitigation strategies to address these risks.

Table 7:Project Risk Analysis

Risk	Impact	Mitigation Strategy
Limited dataset availability	Reduced model accuracy	Use multiple datasets and apply data augmentation

Risk	Impact	Mitigation Strategy
Poor image quality from field conditions	Reduced detection reliability	Apply preprocessing and image normalization
Unknown pest species	Incorrect classification	Integrate Out-of-Distribution detection
Computational limitations	Slow model training	Use cloud-based GPU resources

Appendix E – Ethical Considerations

This appendix outlines the ethical considerations followed during the research. Only publicly available datasets and properly attributed image sources are used for training and testing the machine learning models. No personal or sensitive user data is collected during the development of the system.